SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSENAME: Cloud Computing and Big Data Analytics **COURSECODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct

I N.Rama Sai Kiran Reddy(192525044)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:N.Rama Sai

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data

sovereignty requirements are met? Give an example involving sensitive financial transactions.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS

1. Auto-Scaling and Load Balancing for Seasonal Traffic Spikes

How auto-scaling and load balancing handle the demand

Auto-scaling automatically adds or removes compute instances (VMs or containers) based on rules such as CPU above 70%, request count, or a schedule. A load balancer sits in front of these instances and distributes incoming requests across all healthy servers, so no single server is overwhelmed. Together they let the platform grow during a sale and shrink afterwards, paying only for what is used.

- **Horizontal scaling:** new instances are launched from a pre-built image when load rises (scale-out) and terminated when it falls (scale-in).
- **Health checks:** the load balancer stops sending traffic to failed instances and routes it to healthy ones.
- **Predictive/scheduled scaling:** capacity is pre-warmed before known events such as Black Friday.

Reduced downtime and better user experience

- **No single point of failure:** if one server or availability zone fails, others continue serving requests.
- **Consistent response time:** extra capacity keeps page loads fast, so fewer timeouts, abandoned carts and error pages.
- **Cost efficiency:** no need to over-provision hardware for the whole year.

Role of monitoring tools

Monitoring tools (e.g., Amazon CloudWatch, Azure Monitor, Google Cloud Monitoring, Prometheus/Grafana) collect metrics such as CPU, memory, latency, request rate and error rate. These metrics trigger the auto-scaling policies, raise alerts to the operations team, and provide dashboards and logs for troubleshooting and capacity planning.

Real-world example

An online retailer such as Amazon or Flipkart runs its web tier in an Auto Scaling Group behind an Elastic Load Balancer. During a Big Billion Days sale, traffic rises from 10,000 to 500,000 requests per minute. CloudWatch detects high CPU and latency, and the group scales from 10 to 200 instances within minutes. The load balancer spreads orders evenly, and after the sale the extra instances are removed. Customers experience no slowdown and the company avoids paying for idle servers.

2. Secure and Cost-Effective Cloud Backup Storage

Durability and data redundancy

Cloud object storage (Amazon S3, Azure Blob, Google Cloud Storage) automatically replicates each object across multiple disks, servers and data centres (availability zones). Amazon S3, for example, is designed for 99.999999999% (11 nines) durability. Options such as cross-region replication, versioning and checksums protect against hardware failure, accidental deletion, corruption and regional outages. Lifecycle policies move older backups to archive tiers automatically.

Encryption and access control

- **Encryption in transit:** TLS/HTTPS protects data while it is uploaded or restored.
- **Encryption at rest:** AES-256 with provider-managed keys or customer-managed keys (KMS, HSM) keeps stored data unreadable without the key.
- **Access control:** Identity and Access Management (IAM) policies enforce least privilege, role-based access and multi-factor authentication; bucket policies block public access.
- **Protection from ransomware:** object lock/immutability (WORM) and versioning prevent backups from being altered or deleted.
- **Auditing:** access logs (e.g., CloudTrail) record who accessed or changed data.

Cost advantages over traditional storage

Aspect	Traditional Storage	Cloud Storage
Upfront cost	High (hardware, tapes, data centre space)	None; pay-as-you-go
Scalability	Manual purchase and installation	Virtually unlimited, instant
Maintenance	Own staff, power, cooling, replacement	Handled by provider
Redundancy	Separate DR site needed	Built in across zones/regions
Tiering	Limited	Hot, cool and archive tiers lower the cost of old data

Disaster recovery use case

A hospital stores nightly database backups in cloud storage replicated to a second region. If a fire or ransomware attack destroys the primary data centre, the IT team restores the latest backup into virtual machines in the secondary region and redirects users using DNS failover. Services resume within hours (low RTO) with minimal data loss (low RPO), without having built and maintained a second physical site.

3. Global Deployment Using CDNs and Edge Servers

How CDNs improve performance and reduce latency

A Content Delivery Network is a geographically distributed network of servers that caches content close to users. Instead of every request travelling to the origin server, the user is routed (via DNS or anycast) to the nearest point of presence. This shortens the physical distance, cutting round-trip time, and reduces load and bandwidth cost on the origin.

- **Caching:** static assets such as images, videos, scripts and app updates are served from the cache.
- **Optimisation:** compression, HTTP/2 or HTTP/3, TLS termination and connection reuse speed up delivery.
- **Security:** DDoS absorption and web application firewall at the edge.

How edge servers enhance user experience

Edge servers sit in cities worldwide and deliver content with low latency regardless of region. They can also run lightweight code (e.g., Lambda@Edge, Cloudflare Workers) for authentication, personalisation and API routing, and they absorb traffic spikes so users in Asia, Europe and America all get similar speed and fewer buffering or loading delays.

How cloud providers support global availability

- **Multiple regions and availability zones:** the app is deployed in several regions so users connect to the closest one.
- **Global load balancing and DNS routing:** latency-based or geo-based routing and automatic failover (e.g., Route 53, Traffic Manager).
- **Data replication:** multi-region databases and storage keep data available and consistent.
- **High-speed backbone network:** the provider's private network links regions with low latency.

Real-time streaming / gaming example

Netflix and YouTube store popular videos on edge caches inside ISP networks, so a viewer in Chennai streams from a nearby server in HD without buffering. In online multiplayer games, matchmaking and game servers are deployed in regional data centres and updates or patches are distributed through a CDN. Players connect to the nearest server, giving low ping, fast downloads and a smooth real-time experience.

4. Compliance and Data Security for a Financial Institution

Regulatory compliance and auditing support

Major cloud providers undergo independent audits and hold certifications such as ISO 27001, SOC 1/2/3, PCI DSS, GDPR alignment and HIPAA readiness. They follow a shared responsibility model: the provider secures the infrastructure, while the customer secures its data, configuration and access. Compliance portals (AWS Artifact, Azure Compliance Manager)

supply audit reports, and tools such as AWS Config, Azure Policy and Security Hub continuously check resources against standards.

Role of logging, monitoring and encryption

- **Logging:** audit trails (CloudTrail, Azure Activity Log) record every user and API action, supporting forensic investigation and regulatory audits.
- **Monitoring:** real-time alerts and SIEM/threat-detection services (GuardDuty, Sentinel) identify unusual logins, fraud patterns or data exfiltration.
- **Encryption:** TLS in transit and AES-256 at rest, with customer-managed keys in KMS/HSM, ensure only authorised parties read sensitive data; tokenisation and masking protect card numbers.
- **Access control:** IAM, least privilege, MFA and network isolation (VPC, private endpoints).

Ensuring data sovereignty

- **Region selection:** store and process data only in approved regions or countries (e.g., India for RBI rules).
- **Policy guardrails:** use policies to block creation of resources or replication outside permitted regions.
- **Key control:** keep encryption keys in-country or in customer-held HSMs.
- **Contracts:** use data processing agreements and sovereign or local cloud offerings, and review backup and support-access locations.

Example involving sensitive financial transactions

A bank runs its online payment system in a cloud region located inside the country. Card data is tokenised and encrypted with customer-managed keys, every transaction and administrator action is logged, and monitoring raises an alert when a card is used from two distant locations within minutes. Auditors review the logs and compliance reports to verify PCI DSS and local regulator requirements, and no transaction data leaves the approved region.

5. Containerization for Application Deployment

Containers vs. virtual machines

Feature	Virtual Machine	Container
Architecture	Each VM has its own full guest OS on a hypervisor	Containers share the host OS kernel; only app and dependencies are packaged
Size	Gigabytes	Megabytes
Start-up time	Minutes	Seconds or less
Performance	Hypervisor overhead	Near-native performance

Feature	Virtual Machine	Container
Resource usage	Heavy; duplicate OS per VM	Lightweight; many more per host
Isolation	Strong, hardware-level	Process-level; weaker unless hardened

How orchestration (Kubernetes) simplifies deployment and scaling

- **Automated deployment:** declarative YAML describes the desired state; Kubernetes schedules containers onto nodes.
- **Auto-scaling:** Horizontal Pod Autoscaler adds or removes pods based on load.
- **Self-healing:** failed containers are restarted or rescheduled automatically.
- **Rolling updates and rollbacks:** new versions are released with zero downtime and can be reverted.
- **Service discovery and load balancing:** services find each other and traffic is distributed across pods.
- **Configuration and secrets management:** settings are separated from code.

Portability across cloud environments

A container image packages the application with its libraries and configuration, so it runs the same on a developer laptop, on-premises servers, AWS (EKS), Azure (AKS) or Google Cloud (GKE). This 'build once, run anywhere' approach avoids vendor lock-in, simplifies migration, supports hybrid and multi-cloud strategies, and ensures consistency across development, testing and production.

Example: microservices-based application

An online food-delivery app is split into independent services: User Service, Restaurant/Menu Service, Order Service, Payment Service, Delivery Tracking Service and Notification Service. Each runs in its own container with its own database, communicating through REST APIs or message queues. Kubernetes scales only the Order Service during dinner hours, and a team can update the Payment Service without redeploying the rest of the application, giving faster releases and better fault isolation.